Lecture plan · for the lecturer

Teaching databases and data engineering at university: unit check

Data engineering · University · Unit check

Time	Materials
40 min	Projector or board, and the lecture slides.

Learning goals

- Normalisation
- Indexing
- Transactions and CAP
- Data quality

Activity: Peer-instruction review

1. Vote (10 min): Show each question; students answer alone and commit to an answer.
2. Discuss (15 min): Where answers split, pairs argue for two minutes and vote again.
3. Explain (10 min): Work through the questions most students missed, showing the reasoning step by step.
4. Plan (5 min): Note which misconception to revisit next week and how.

Discussion

- Which misconception will you target first?
- Which lab will you add?
- Which real dataset will you use?

Exit ticket

- One change to my database course this term, and how I will know it worked.

Try it in your next lecture
Bring your real-data project brief and rubric to your next department meeting for review.

Seminar sheet · for students

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Name _____ Class _____ Date _____

1 **R(A, B, C) with A → B and B → C, key A. The highest normal form is**

◯ 2NF ◯ 3NF ◯ BCNF

2 **Compute the closure.**

With A → B and B → C, $\{A\}^+ = \{A, B, _____\}$.

Word bank: C · D · ∅

3 **An index on a column always makes queries on it faster.**

◯ True ◯ False

4 **A B+-tree with fan-out 100 over 10^6 keys has about**

◯ 3 levels ◯ 6 levels ◯ 20 levels

5 **An index on (dept, year) helps which? Choose all that apply.**

☐ dept = 'CS' ☐ dept = 'CS' AND year = 2 ☐ year = 2